Predicting diabetes risk using health survey data

Executive Summary

This project develops a machine learning model to identify individuals at risk of diabetes or prediabetes using non-invasive health survey data. Diabetes is a major public health concern in the United States, and early detection can help prevent complications and reduce healthcare costs. The goal of this analysis is to determine whether demographic, behavioral, and cardiovascular factors can be used to predict diabetes risk.

The dataset comes from the 2015 Behavioral Risk Factor Surveillance System (BRFSS), which contains about 250,000 survey responses. Important predictors include BMI, age category, high blood pressure, cholesterol status, physical activity, smoking behavior, and general health. Table 1 provides a brief data dictionary describing the main variables used in the analysis.

Table 1. Data dictionary

Variable	**Description**
Diabetes_012	Diabetes status (0 = none, 1 = prediabetes, 2 = diabetes)
HighBP	High blood pressure indicator
HighChol	High cholesterol indicator
BMI	Body Mass Index
Smoker	Smoking status
Stroke	History of stroke

Variable	Description
HeartDiseaseorAttack	History of heart disease or heart attack
PhysActivity	Physical activity indicator
Fruits	Regular fruit consumption
Veggies	Regular vegetable consumption
Sex	Gender indicator
Age	Age category
Education	Education level
Income	Income category

The original three-category diabetes variable was converted into a binary outcome to simplify the prediction task.

Three models were trained and compared: Logistic Regression, Random Forest, and Gradient Boosting. Model performance was evaluated using Accuracy, F1 Score, and ROC AUC. The ensemble models performed better than the linear baseline, with Gradient Boosting providing the strongest overall performance. The most influential predictors included BMI, high blood pressure, age, cholesterol status, and general health.

Overall, the results show that survey-based health indicators can help identify individuals who may be at higher risk for diabetes. While the model is not intended to replace medical diagnosis, it could help healthcare organizations prioritize early screening and prevention efforts.

Introduction & Problem Definition

Diabetes is a chronic metabolic disease that affects millions of people worldwide and is associated with serious complications such as cardiovascular disease, kidney failure, and nerve damage. The increasing prevalence of diabetes creates significant challenges for healthcare systems, both medically and economically. Identifying individuals at high risk early can help support prevention efforts and improve long-term health outcomes.

This project uses data from the 2015 Behavioral Risk Factor Surveillance System (BRFSS), a large national health survey that includes demographic, behavioral, and health-related information such as BMI, blood pressure, cholesterol status, age, smoking behavior, and general health. The main research question is whether diabetes or prediabetes risk can be predicted using these survey-based health indicators.

The problem is formulated as a supervised binary classification task. The original three-category diabetes variable was converted into a binary outcome (0 = no diabetes; 1 = prediabetes or diabetes). Model performance is evaluated using ROC AUC as the primary metric, along with Accuracy and F1 Score. The objective is to build a reliable and interpretable model that can help support diabetes risk screening while recognizing that it is not intended to replace clinical diagnosis.

Data Overview

The dataset used in this project comes from the 2015 Behavioral Risk Factor Surveillance System (BRFSS), a nationwide health survey conducted by the Centers for Disease Control and Prevention (CDC). It contains about 250,000 observations and more than 20 demographic, behavioral, and health-related variables, including BMI, blood pressure, cholesterol status, age, smoking behavior, and general health.

The original diabetes variable included three categories (no diabetes, prediabetes, diabetes). For this analysis, it was converted into a binary outcome: 0 = No diabetes, 1 = Prediabetes or diabetes.

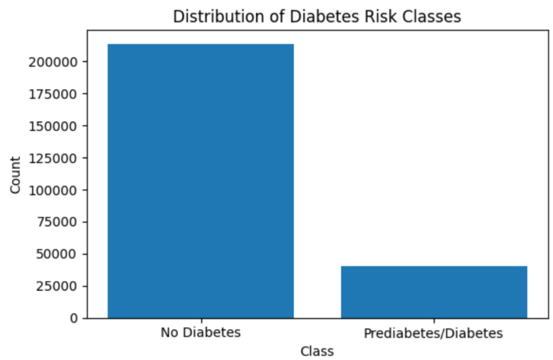

Figure 1. Distribution of the binary diabetes outcome.

This simplifies the task into a binary classification problem focused on identifying individuals at higher risk of diabetes.

Data Quality and Ethical Considerations

The dataset was generally clean, with minimal missing values handled using median imputation. Because the data is self-reported and does not include clinical biomarkers, some limitations exist.

Since the dataset includes demographic and socioeconomic information, the model should be used carefully as a risk screening tool rather than a medical diagnosis, with attention to fairness across populations.

Data Cleaning and Preprocessing Methods:

The dataset required minimal cleaning because most variables were already numeric or category coded. Missing values were limited and handled using median imputation for numerical variables. No rows were removed due to the low rate of missing data.

Item	**Value**
Observations	~253,000
Features	21
Target	Diabetes_binary
Missing values	minimal

Outliers were examined using summary statistics and boxplots. Some high BMI values were observed but were retained because they represent plausible real-world cases.

The original three-category diabetes variable was converted into a binary to simplify the classification task. Features for Logistic Regression were standardized using standard scaling, while tree-based models were trained without scaling. No dimensionality reduction was applied to maintain interpretability.

Exploratory Data Analysis:

Exploratory Data Analysis (EDA) was conducted to examine variable distributions and their relationships with diabetes status. The target variable showed a moderate class imbalance, approximately **84% of observations belong to the non-diabetes class, while about 16% represent individuals with prediabetes or diabetes**, confirming the presence of class imbalance in the dataset. which justified the use of class weighting in Logistic Regression.

BMI showed clear separation between individuals with and without diabetes risk, suggesting strong predictive value. Diabetes prevalence also increased across age categories. Cardiovascular factors such as high blood pressure and high cholesterol were strongly associated with diabetes, and individuals reporting poorer general health had higher risk.

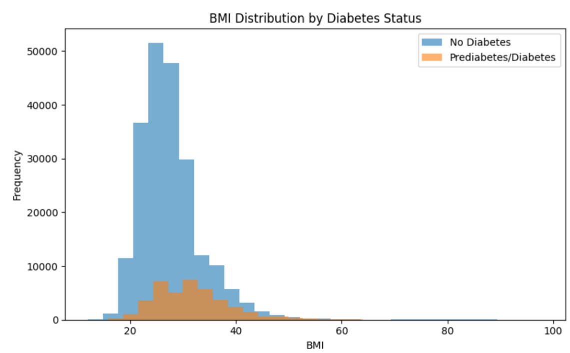

Figure 2. BMI distribution by diabetes status.

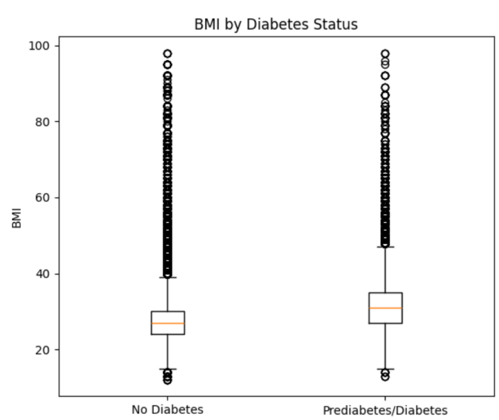

Figure 3. BMI comparison between diabetes groups.

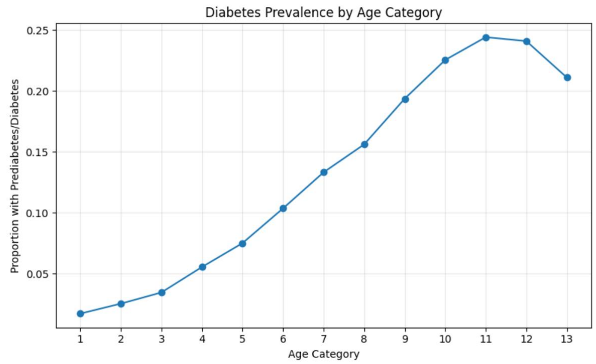

Figure 4. Diabetes prevalence across age categories.

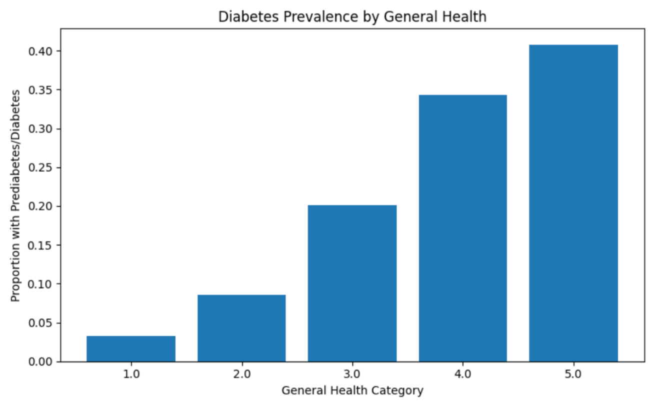

Figure 5. Diabetes prevalence by general health status.

Correlation analysis confirmed positive relationships between BMI, cardiovascular indicators, and diabetes status. No severe multicollinearity issues were observed.

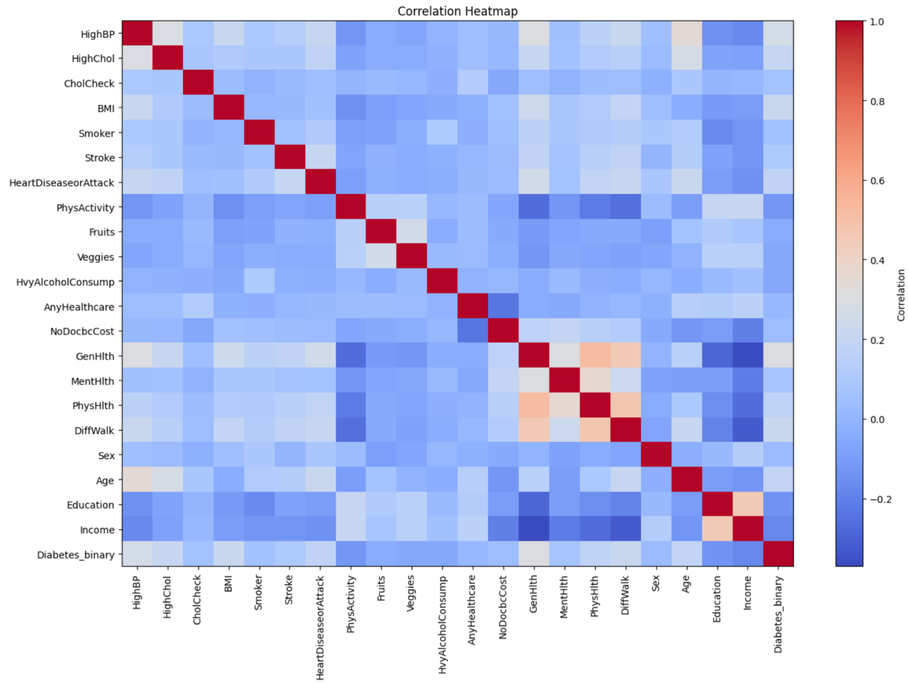

Figure 6. Correlation heatmap of key variables.

Overall, the dataset appeared clean and well-structured, and the EDA results supported the relevance of key health risk factors for predicting diabetes.

Modeling / Analysis

This project is formulated as a supervised binary classification problem. The goal is to predict whether an individual has prediabetes or diabetes (1) versus no diabetes (0) using demographic, behavioral, and cardiovascular indicators. Because the outcome variable is labeled and binary, classification algorithms are appropriate.

The objective is to compare linear and nonlinear models to determine whether more complex relationships between predictors improve diabetes risk prediction.

Model Selection

Three models were implemented with increasing levels of complexity.

Logistic Regression was used as a baseline linear classifier because it is interpretable and provides a useful benchmark. Class weighting was applied to address moderate class imbalance. Random Forest was used to capture nonlinear relationships and feature interactions. As a bagging-based ensemble method, it reduces variance and performs well with structured tabular data. Gradient Boosting was included as a boosting-based ensemble model that iteratively improves predictions by correcting previous errors. It is widely known for strong performance in tabular machine learning tasks, **Table 2 summarizes the performance of the three classification models.**

Table 2. Model Performance Comparison

Model	**Accuracy**	**F1 Score**	**ROC AUC**
Gradient Boosting	0.851	0.299	0.824
Logistic Regression	0.730	0.471	0.818
Random Forest	0.845	0.303	0.794

Together, these models allow comparison between a linear baseline and more flexible ensemble approaches.

Training Process and Data Splitting

The dataset was split into 80% training and 20% testing data using stratified sampling to preserve the class distribution. Models were trained on the training data and evaluated on the test set to assess generalization performance.

Features for Logistic Regression were standardized using z-score scaling, while tree-based models were trained without scaling due to their scale invariance. Cross-validation was considered, but a stratified holdout split was sufficient given the large dataset size (~250,000 observations).

Evaluation Metrics

Model performance was evaluated using three metrics:

- Accuracy – overall classification performance
- F1 Score – balances precision and recall
- ROC AUC – primary evaluation metric

ROC AUC was selected as the main success criterion because it measures how well the model separates individuals with and without diabetes risk across different classification thresholds.

Hyperparameter Tuning

Basic hyperparameter tuning was performed to improve performance:

- Logistic Regression: increased iterations and applied class weighting
- Random Forest: adjusted the number of trees
- Gradient Boosting: tuned the learning rate and number of estimators

All models were evaluated using the same training and testing split to ensure a fair comparison.

Results:

All three models—Logistic Regression, Random Forest, and Gradient Boosting—were evaluated using Accuracy, F1 Score, and ROC AUC. ROC AUC was used as the primary performance metric because it measures how well a model distinguishes between individuals with and without diabetes risk.

The results show that the ensemble models performed better than Logistic Regression. While Logistic Regression provided a useful baseline, Random Forest and Gradient Boosting achieved higher ROC AUC values, indicating stronger ability to distinguish between high-risk and low-risk individuals. Gradient Boosting performed best likely because boosting methods iteratively reduce prediction errors and are able to capture complex nonlinear relationships between health indicators and diabetes risk. The relatively low F1 scores are influenced by the class imbalance in the dataset, where the majority class (no diabetes) is much more common than the positive class. This suggests that nonlinear relationships among health variables play an important role in predicting diabetes risk.

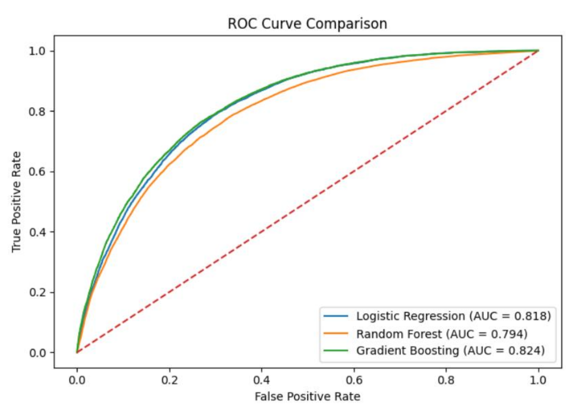

Figure 7. ROC curve comparison for the three models.

Feature importance analysis from the tree-based models identified the most influential predictors as BMI, high blood pressure, age category, high cholesterol, and general health. These variables are widely recognized medical risk factors for diabetes, which supports the reliability of the model results.

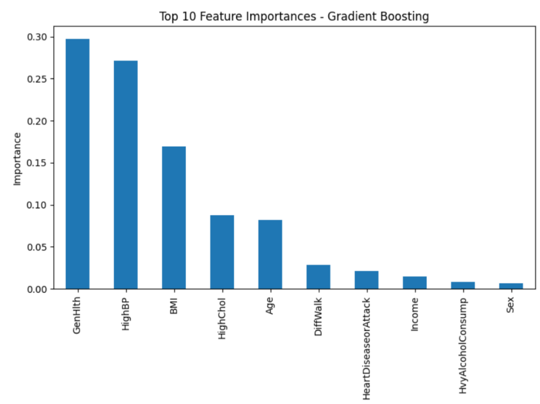

Figure 8. Feature importance for the Gradient Boosting model.

The ROC curves show strong separation between the two classes, remaining well above the diagonal line that represents random guessing. The confusion matrix also indicates that the model correctly classified most individuals, although some false positives and false negatives remain.

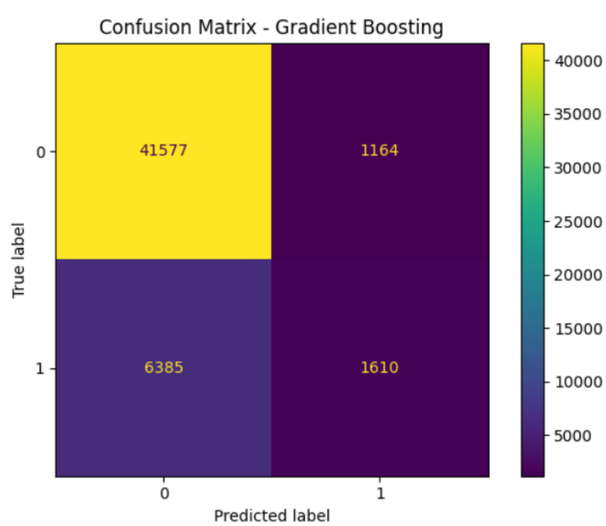

Figure 9. Confusion matrix for the Gradient Boosting model.

There are several limitations to consider. The dataset is based on self-reported survey data and does not include clinical measurements such as blood glucose levels. In addition, the model reflects patterns from a single survey year and may not fully generalize to other populations. Fairness across demographic groups was also not deeply analyzed and would require further investigation.

Despite these limitations, the results suggest that survey-based health data can help identify individuals at higher risk for diabetes. Healthcare organizations could use similar models to support early screening and prevention programs. However, the model should be used only as a risk assessment tool rather than a medical diagnosis.

Recommendations:

Based on the results of this project, the model can be used as a **risk screening tool** to help identify individuals who may be at higher risk of diabetes. While it should not be used as a medical diagnosis, it can help healthcare organizations determine which individuals may benefit from further testing or preventive support.

What the Client Should Do

Healthcare providers or public health organizations could use this model to:

- Identify individuals with **high BMI, high blood pressure, and poor general health** who may be at higher risk.
- Invite high-risk individuals for **additional clinical testing**.
- Provide **lifestyle education programs** focused on diet, exercise, and weight management.

- Prioritize **older populations and individuals with cardiovascular risk factors** in prevention programs.

Using this model can help organizations focus their resources on individuals who are most likely to benefit from early intervention.

Implementation Considerations

Before applying the model in practice, several steps should be considered:

- Validate the model using **newer or external datasets**.
- Evaluate performance across **different demographic groups** to ensure fairness.
- Clearly communicate that the model is intended for **risk screening, not medical diagnosis**.
- Monitor model performance over time and **update it when necessary**.

Future Improvements

This project could be improved by incorporating additional data and modeling approaches, such as:

1. Including **clinical biomarkers** such as blood glucose measurements.
2. Applying **cross-validation and more advanced hyperparameter tuning**.
3. Analyzing **multiple years of survey data** to examine how diabetes risk changes over time.

Overall, the results suggest that survey-based health data can help identify individuals at higher risk for diabetes and support **early prevention strategies in public health settings**.

Conclusion:

This project examined whether diabetes or prediabetes can be predicted using survey-based health and lifestyle data. The analysis included data exploration, model development, and comparison of multiple machine learning approaches.

The results show that diabetes risk can be predicted with strong performance using commonly collected health indicators. Ensemble models such as Random Forest and Gradient Boosting performed better than Logistic Regression, suggesting that nonlinear relationships between risk factors play an important role. Key predictors included BMI, high blood pressure, age, cholesterol status, and general health, which are consistent with known medical risk factors.

However, several limitations should be considered. The dataset is based on self-reported information and does not include clinical biomarkers such as blood glucose measurements. In addition, the data represents a single survey year and may not fully generalize to other populations.

Despite these limitations, the findings demonstrate that machine learning can transform large public health datasets into useful insights for identifying individuals at higher risk of diabetes. While the model should not be used as a medical diagnosis, it can support early screening strategies and preventative healthcare planning. Future work could improve the model by incorporating clinical data, validating results on additional datasets, and evaluating fairness across demographic groups.

These findings highlight the potential value of data-driven approaches for improving early disease detection and supporting preventive public health strategies.

AI Appendix:

ChatGPT (OpenAI) was used to help organize the structure of this report and improve grammar and clarity in some sections. The AI tool was mainly used to review written paragraphs and suggest clearer wording.

Example prompts used include:

- "Make this paragraph shorter and more natural."
- "Improve the grammar of this section."

All data analysis, coding, and interpretation of the results were completed by me.

Data Sources

Centers for Disease Control and Prevention. (2015). *Behavioral Risk Factor Surveillance System (BRFSS) annual data 2015*. https://www.cdc.gov/brfss/annual_data/annual_2015.html

Kaggle. (n.d.). *Diabetes health indicators dataset*. https://www.kaggle.com/datasets/alexteboul/diabetes-health-indicators-dataset